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A Regime-Switching Framework for Tactical Sector Tilts under Higher-Moment Investor Preferences

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Abstract

This paper develops and evaluates a regime-aware core-plus-satellite allocation framework designed for implementation within the operational constraints of a wealth management context. A three-state Hidden Markov Model is estimated on monthly excess log returns of the S&P 500 Total Return index and the Bloomberg US Treasury 7–10 Year index. The resulting one-step-ahead predictive regime probabilities serve as the signal for a scoring engine that evaluates nine US sector ETFs and gold as candidate tilts against a 60/40 benchmark. Tilt scoring follows a Taylor-expanded expected utility function after Jondeau and Rockinger (2006), accommodating mean-variance, mean-variance-skewness, and mean-variance-kurtosis investor types, and a rolling 60-month estimation window allows the model to adapt to structural breaks in sector leadership. A score improvement floor holds the portfolio in the passive core in any month in which no candidate tilt improves on it by more than the floor. Three further mechanisms target drawdown protection, all confined to the sizing stage and none visible to selection: a style-dependent displacement rule funds defensive tilts from the equity leg alone, leaving the bond block intact in risk-off months; a downside realised-semivariance ratio computed from daily returns sharpens the bear probability that sizes the satellite sleeve; and a blended bond–equity correlation sets the internal equity–bond split of the core. At a 30% satellite sleeve the framework achieves a CAGR of 9.26% against a benchmark of 7.74%, a Sharpe ratio of 1.029 versus 0.874, a maximum drawdown of -26.6% versus -30.6% , and a downside deviation of 6.99% versus 7.07% — improving on the benchmark on every risk-adjusted metric simultaneously. A 35% sleeve raises CAGR to 9.64% at a Sharpe ratio of 1.039. At both operating points up-capture exceeds 102% of the benchmark while down-capture remains below 90%.

Keywords: Hidden Markov Model, regime switching, tactical asset allocation, sector rotation, higher-moment preferences, rolling window estimation, core-plus-satellite portfolio

JEL Classification: G11 (Portfolio Choice; Investment Decisions) · G17 (Financial Forecasting and Simulation) · C22 (Time-Series Models) · C61 (Optimization Techniques) · G23 (Pension Funds; Other Private Financial Institutions)

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1. Introduction

In the wealth management business, only a small percentage of clients are managed actively, and that depends largely on the size and the cost of continuing operations of the firm. Most individual clients are managed passively or on very infrequent rebalancing schedules, and their portfolios are standardised to facilitate the communication chain between client, relationship manager, portfolio manager, and trading desk, and to abide by rules and regulations.

Once a standardised approach has been developed for a firm, an operational pipeline follows, and its scale depends, again, on the size and cost of operations. According to PwC's Asset and Wealth Management 2025 report, passive AUM is projected to reach USD 70 trillion by 2030 at an annualised rate of 10%, and 89% of asset managers reported profitability pressure over the five years preceding the report, with profit per AUM declining by approximately 19% since 2018. [PV: your note asked whether to add more sources here. My view: one more is enough and the trend is already stark. Suggest adding a single fee-compression datapoint (e.g. average expense ratios) and stopping there — in 5–7 pages this paragraph earns its place by being short and pointed. Tell me if you want me to draft the sentence.] This is the operational economics argument that motivates the study: under cost pressure and growing AUM, the practical default for smaller clients is a standardised passive mandate.

This poses the question at the core of this study: is it possible to develop a model with a decision tilt that outperforms a passive benchmark on a risk-adjusted basis and that simultaneously addresses the concerns of investors with different preferences for higher moments? Most studies that use a regime switching framework to solve a portfolio allocation problem do so considering different types of investors, but not the type of portfolio manager who operates within a standardised, constrained mandate. This study deliberately caters to that audience.

The main empirical results are reported in **Table 7**. We contribute to the regime-switching and portfolio allocation literature in four ways. First, the HMM is refit at every rebalance date within a rolling 60-month window, allowing the model to adapt to structural breaks in sector leadership. Second, allocation decisions are scored rather than optimised, making the approach computationally manageable within a real operational pipeline. Third, multiple investor preference types, from mean-variance to mean-variance-kurtosis, are accommodated within the same framework. Finally, a score improvement floor acts as a hurdle rate, so the model deviates from the passive core only when the regime signal justifies it.

These four elements define the selection engine, which scores candidate tilts on the predictive moments of the regime mixture alone. We extend that engine with three mechanisms aimed at drawdown protection, all confined to the sizing stage: a style-dependent displacement rule that funds defensive tilts from equity so the bond block survives risk-off months; a downside realised-semivariance signal from daily returns that sharpens the bear probability sizing the

sleeve; and a blended bond–equity correlation that sets the internal equity–bond split of the core. Keeping all three out of selection preserves the two-stage separation: the regime model decides *what* to hold and *which way* to lean, the scalars only *how much*.

[cut: intro paragraph on Ang & Bekaert / Guidolin & Timmermann — duplicated verbatim in Section 2.1, so removed here; the PV note on adding a fourth departure is retained below]

The present study sits in that tradition but departs from it in three ways: a simpler, operationally manageable scoring approach rather than full Bellman equation optimisation; a US sector ETF universe rather than international equity; and rolling window re-estimation as the mechanism for handling structural breaks, which studies relying on the stable 1954–1999 data environment did not require.

The thesis is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents the Hidden Markov Model specification and estimation. Section 5 develops the investor preference theory and scoring function. Section 6 describes the allocation layer architecture. Section 7 presents the satellite tilts, conviction scalar, and rolling window mechanics. Section 8 reports the key findings. Section 9 discusses gold as a conditional safe haven. Section 10 concludes. [PV: roadmap to be rewritten once the final section structure is fixed. Holding until the 5–7 page skeleton is agreed — no point renumbering twice.]

2. Literature Review

[PV: you are rewriting this section with the volatility and correlation literature. The hybrid-B paragraph is now in the Sector ETF subsection below (with Grinold & Kahn). For the new material the load-bearing citations are Connolly, Stivers & Sun (2005) for the armed gate, Moreira & Muir (2017) and Patton & Sheppard (2015) for the volatility signal, Baele, Bekaert & Inghelbrecht (2010) for the correlation horizon, and Cederburg et al. (2020) as the counterpoint.]

2.1 Regime Switching in Asset Allocation

The regime-switching framework originates with Hamilton (1989), whose filtered probability recursion allows the latent state to be inferred from observed data, producing a probabilistic assessment of the current regime rather than a binary classification. The first direct application to portfolio allocation is Ang and Bekaert (2002), who show that correlations increase substantially during bear regimes, eroding the diversification benefit static mean-variance analysis predicts. Ang and Bekaert (2004) demonstrate that regime-aware allocation outperforms a static 60/40 benchmark even after transaction costs, with the key mechanism being regime persistence: regimes last long enough for the signal to be actionable. Guidolin and Timmermann (2007) provide the most comprehensive treatment, estimating a four-state model on US stocks and bonds and solving the full dynamic programming problem; the welfare cost of ignoring regimes reaches approximately eight cents per dollar at a 60-month horizon under high risk aversion. A finding directly relevant here is that no VAR terms are required, suggesting the return distribution itself, rather than predictive variables, carries the information. The Ang and Timmermann (2012) survey documents that three or more states outperform two-state models, supporting the specification used here.

[cut: 2.1: Hamilton EM detail, the all-equity welfare-cost figures, G&T state names and out-of-sample discussion, and the three-contributions paragraph — the last is stated in the Introduction]

2.2 Sector ETF Allocation under Regime Switching

The application of regime switching to sector-level allocation is a narrower and more recent strand. Ma, MacLean, Xu, and Zhao (2011) is the closest antecedent, proposing a regime-dependent factor-risk-neutral strategy for the nine State Street SPDR sector ETFs, the same universe used here and selected on the same criteria: a single provider, a long trading history beginning December 1998, and comprehensive coverage of the US market. Their framework does not include a fixed bond core, does not incorporate higher-moment preferences, and uses an expanding estimation window; the rolling window adaptation and the core-plus-satellite architecture are contributions of the present study. Dou, Gallagher, Schneider, and Walter (2014) provide direct evidence that cross-sector regime-dependent allocation outper-

forms cross-regional allocation, reaching an out-of-sample Sharpe ratio of 0.93 against 0.30 for the world market portfolio, with notably lower beta in bear regimes. Their documentation of lower cross-sector correlation during bear regimes supports the motivation for the conviction scalar used here. Jiang, Liu, and Tse (2015) find comparable results on international iShares ETFs after transaction costs, evidence that ETF-based regime-switching strategies are implementable at the institutional level. [cut: 2.2: benchmark-constrained Sharpe figures from Dou et al., and the daily-rebalancing detail from Jiang et al.]

[PV: the two paragraphs below are mine and you said you want to rewrite them in your own voice — left untouched for that. They are currently 250 words; budget 120.]

A dimension this literature leaves unexamined is the *funding* of a sector tilt: which existing holdings are sold to pay for it. It matters because the two satellite styles play opposite roles. A cyclical tilt amplifies equity risk and is naturally financed by the portfolio as a whole; a defensive tilt substitutes away from equity risk, and financing it pro-rata would liquidate part of the bond block at exactly the moment that block is most valuable. The present study therefore funds cyclical satellites pro-rata from the core and defensive satellites from the equity leg alone, so the bond weight scales only with the cyclical allocation. Following the separation advocated by Grinold and Kahn (2000) between the model that ranks opportunities and the rules that implement them, this is a portfolio-construction rule rather than an alpha signal: it is applied identically to every candidate and uses only the static style label of each satellite, so it never enters the ranking itself.

This funding rule shares its motivation with the two conditioning scalars introduced in Section 6. The upside of the regime signal was already established in the thesis-stage results; what remained was protection against sudden drawdown episodes, and all three mechanisms are organised around the same bear-regime read. The displacement rule preserves the bond block structurally whenever the model turns defensive; the volatility scalar sharpens the bear probability that triggers that rotation; and the correlation scalar decides how much of the core should be held in bonds once it does. Because the three act on different parts of the portfolio — funding, sleeve size, and core composition — but respond to the same underlying state, they compound rather than offset, which is why the downside metrics improve materially while the upside capture is preserved.

2.3 Higher-Moment Preferences and Taylor-Expanded Utility

Scott and Horvath (1980) provide the theoretical foundation: their result that investors prefer positive skewness and dislike excess kurtosis holds broadly for risk-averse utility functions rather than specific parametric forms, so the investor types defined here rest on a universal property of risk-averse behaviour rather than an assumed functional form. Jondeau and Rockinger (2006) supply the operational implementation, a Taylor expansion of expected utility around the moments of the return distribution that yields a closed-form scoring func-

tion and avoids the computationally prohibitive co-skewness and co-kurtosis tensors direct optimisation would require. They find the opportunity cost of using only mean and variance ranges from negligible in near-normal environments to more than 25 cents per dollar in highly non-normal settings. Where they evaluate a single combined MVSK investor, we separate MVS and MVK types so the contribution of each moment can be isolated empirically.

[cut: 2.3: Kim, Fabozzi, Cheridito & Fox (2014) on robust mean-variance implicitly favouring skewness — complementary but not load-bearing]

2.4 Rolling Window Estimation and Time-Varying Parameters

The rolling window approach is grounded in the parameter instability literature. Pesaran and Timmermann (2007) establish that under structural breaks the optimal window trades off bias from stale data against variance from insufficient observations, and that rolling windows are preferred when the data-generating process exhibits parameter instability. Hematizadeh, Tajaddini, and Hallahan (2022) find that even constant-transition-probability HMMs with rolling estimation capture most of the time-variation benefit of more complex TVTP models, supporting the simpler constant- P specification used here. Nystrup, Hansen, Madsen, and Lindström (2015) show that regime-based allocation with adaptive parameter estimation outperforms static allocation without requiring return forecasting skill, and their use of a confidence threshold before changing positions is the conceptual antecedent of the score improvement floor.

[cut: 2.4: the $k = 3$ design-tension paragraph. It is a LIMITATION rather than literature — suggest one sentence in the robustness section instead: k is selected on the full sample, so it is a modelling assumption rather than a fully out-of-sample choice]

[cut: 2.4: the extended discussion of why re-selecting k each window would be computationally expensive and methodologically inconsistent — reduced to the caveat]

2.5 Volatility and Correlation as Conditioning Signals

Numerous studies, most notably Moreira and Muir (2017) and Patton and Sheppard (2015), show empirically that high volatility and volatility spikes tend to be followed by further elevated volatility, and to be preceded by negative returns. Patton and Sheppard show that this asymmetry is driven specifically by the downside component of realised variance, which is why the signal used here is a realised semivariance rather than a total variance.

A broad literature explores the time-varying relation between assets, and in particular the stock–bond relation. The sign of that relation is not stable: during crises and drawdown episodes the correlation between broad stocks and bonds flips negative, signifying flight-to-quality market movements. At the same time, Baele, Bekaert and Inghelbrecht (2010) find that macroeconomic fundamentals play little role in determining or even explaining stock–

bond correlation. Their study underlines one of the reasons why we chose to keep this model endogenous and explore only volatility and correlation as potential scaling signal candidates.

In *Modeling and Forecasting Realized Volatility*, Andersen, Bollerslev, Diebold and Labys (2003) show that high-frequency data can be aggregated to measure volatility accurately at lower frequencies. Their work uses exchange rates, but the authors underline the parallels to other asset classes. When we researched how monthly rebalancing could be influenced by volatility and correlation signals of higher frequency, a major concern was that daily data would introduce too much noise into the model. For that reason, we left the daily volatility and correlation data to work as a separate mechanism in the architecture, measuring the state of a month rather than triggering trades within it.

Their second main argument is that parametric volatility modelling becomes unwieldy as the number of assets grows, whereas realised measures remain a model-free and essentially unbiased estimator of return volatility, and stay tractable in a high-dimensional setting. Our model explores a large number of candidate combinations at each monthly rebalancing date, so this tractability is a practical requirement rather than a theoretical nicety. Following Andersen et al., we build a non-conflicting volatility and correlation mechanism.

Focusing on correlation, Ledoit and Wolf (2004) argue that covariance matrices contain substantial estimation error, and propose a shrinkage mechanism that rectifies it. The central idea is that extreme covariance values are pulled towards the centre, much as a winsorised mean pulls in outliers. We apply the same idea to the estimation error produced by small regimes: whenever a regime holds fewer than twelve observations inside the current five-year window, its covariance is replaced by the pooled window covariance, so that the slow correlation layer is never represented by an unstable estimate. On top of that we blend it with a faster-reacting correlation layer that further dilutes any erroneous signalling into the scaling mechanism.

[PV: two checks. (i) Your draft said Andersen et al. show “worse results when forecasting daily” — as I read them, their realised-volatility forecasts beat GARCH at both one- and ten-day horizons, so I left the claim out. Verify before restoring. (ii) Your “over long periods the relation stays low positive” holds pre-2000 but not 2000–2020, so I cut that sentence rather than assert it. (iii) I replaced the direct quotation from Andersen et al. with a paraphrase.]

3. Data

In order to define the regimes, one broad equity index and one broad bond index were used: the S&P 500 total return index and the Bloomberg US Treasury 7–10 Year index (LT09TRUU). These monthly returns were continuously compounded. The S&P 500 total return series was obtained through the Yahoo Finance API, and the Bloomberg Treasury index through Bloomberg. The 1-month T-Bill rate was subtracted from both to calculate the excess log return, which is how both series are referred to in this study. The monthly T-Bill simple return was acquired through the official Fama-French website and converted to log return for comparability with the other data. The start of the sample period for constructing the HMM is January 1999 and the end date is March 2026, matched to the issuance dates of the sector ETFs used for satellite tilts.

For the satellite universe, this study uses the nine State Street SPDR sector ETFs: Materials (XLB), Energy (XLE), Financials (XLF), Industrials (XLI), Technology (XLK), Consumer Staples (XLP), Utilities (XLU), Health Care (XLV), and Consumer Discretionary (XLY). Gold (XAU) is included as a tenth satellite candidate. All nine sector ETF return series were obtained through the Yahoo Finance API. The gold (XAU) return series was obtained from Bloomberg.

The selection of LT09TRUU as the bond component of the regime identification model was made after comparing three bond index variants: the Bloomberg US Aggregate Bond Index (LBUSTRUU), the Bloomberg US Treasury 7–10 Year index (LT09TRUU), and a mutual fund proxy tracking the Aggregate index (WFBIX). The Treasury-only index produced the most economically interpretable and stable regimes, particularly across the 2001 dotcom crash, the 2008 Global Financial Crisis, and the 2020 COVID-19 period. This is consistent with the intuition that a pure-rate-risk instrument with no credit exposure provides a cleaner signal for the bond-equity correlation dynamics that drive regime identification.

During the exploration phase of this project, monthly data from the EURUSD exchange rate, WTI crude oil, and bond indices at alternative durations were also tested with the Hidden Markov Model. Those outputs were either too noisy or could not be realistically interpreted in a macroeconomic context. Given the focus of the study on sector ETF tilts, their inclusion could not be reasonably justified at this stage of the research.

Figure 1 shows the full price history of all twelve assets. Together, the S&P 500 TR and LT09TRUU constitute the 60/40 benchmark core at a fixed weight allocation. The choice of a 60/40 split is the industry standard for a balanced client mandate and represents the passive baseline against which all strategy configurations are evaluated.

Rebalancing timeline and data construction. All series are monthly end-of-month log returns from total-return adjusted prices. Bloomberg indices are total-return series. The

no-look-ahead convention is strict: the HMM is estimated using returns through month t ; predictive probabilities $\pi_{t+1|t}$ are computed from the filtered state at t ; weights are set at the close of month t and earn returns in month $t + 1$. Transaction costs of 5 basis points per side are deducted at rebalancing. The risk-free rate is aligned to the same month-end.

Figure 1 — Cumulative Price Levels of Core and Satellite Assets (Feb 2004 = 100)

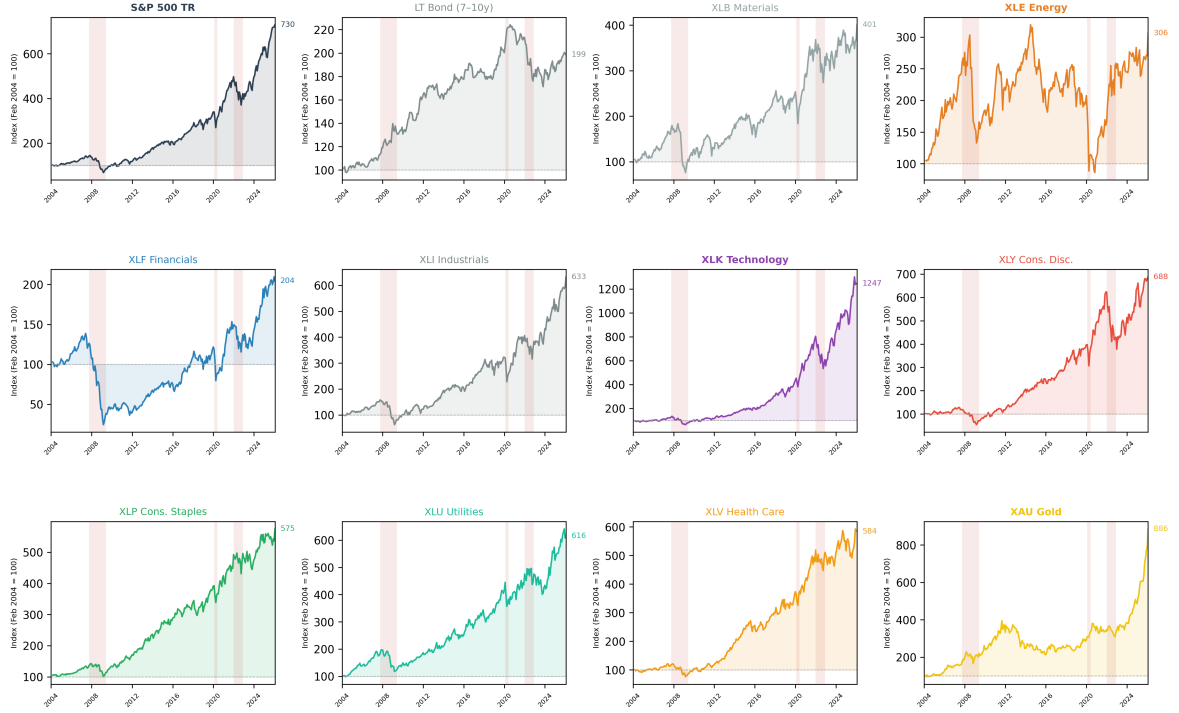


Figure 1: Cumulative price levels of the core and satellite assets, rebased to 100 at February 2004. Shaded regions mark the four crisis episodes. Final index values are annotated at the right of each panel. XLK (Technology, 1247) and XAU (Gold, 896) are the highest-returning satellites over the full period; XLE (Energy, 306) and XLF (Financials, 204) are the weakest cyclical performers, reflecting the post-2014 energy bear market and the persistent post-GFC underperformance of financials respectively.

Table 1: Descriptive Statistics for Monthly Log Returns — Full Sample (Feb 2004 – Jan 2026)

Asset	Ann. Mean (%)	Ann. Vol (%)	Sharpe	Skewness	Exc. Kurt.	Min (%)	Max (%)
<i>Core assets</i>							
S&P 500 TR	10.16	14.65	0.523	−0.806	1.831	−18.39	12.06
LT Bond (7–10y)	3.33	6.41	0.129	0.056	0.868	−4.80	7.69
<i>Cyclical sector satellites</i>							
XLB Materials	8.32	19.77	0.294	−0.567	1.842	−25.35	15.99
XLE Energy	8.63	25.99	0.236	−0.667	4.539	−42.12	26.82
XLF Financials	5.72	21.67	0.149	−1.120	4.116	−30.38	19.72
XLI Industrials	10.12	18.30	0.417	−0.733	2.334	−20.61	16.61
XLK Technology	13.14	17.92	0.594	−0.485	0.376	−17.58	12.87
XLY Cons. Disc.	10.52	18.53	0.433	−0.257	1.415	−19.40	17.30
<i>Defensive sector satellites</i>							
XLP Cons. Staples	8.66	11.69	0.527	−0.578	1.168	−13.48	9.94
XLU Utilities	9.31	14.21	0.479	−0.848	1.020	−13.93	10.03
XLV Health Care	8.98	13.62	0.476	−0.379	0.449	−13.23	11.86
<i>Commodity satellite</i>							
XAU Gold	11.36	16.75	0.529	−0.184	0.621	−18.50	12.49

Notes: Monthly log returns from the allocation backtest period (February 2004 – January 2026, $T = 264$ monthly observations). Annual mean and volatility obtained by multiplying monthly mean by 12 and monthly standard deviation by $\sqrt{12}$. Sharpe ratio uses an approximate average 1-month T-bill rate of 2.5% per annum as the risk-free rate. Skewness and excess kurtosis use Fisher’s definition. All nine sector ETF returns obtained via Yahoo Finance API; XAU obtained from Bloomberg; LT Bond = LT09TRUU Bloomberg US Treasury 7–10 Year index.

Table 2: Pairwise Correlation Matrix of Monthly Log Returns (Feb 2004 – Jan 2026)

	SP500	Bond	XLB	XLE	XLF	XLI	XLK	XLY	XLP	XLU	XLV	XAU
SP500	1.00	−0.06	0.86	0.63	0.85	0.92	0.90	0.90	0.72	0.51	0.75	0.09
Bond		1.00	−0.11	−0.25	−0.16	−0.11	−0.03	−0.03	0.10	0.22	0.04	0.32
XLB			1.00	0.68	0.75	0.87	0.71	0.77	0.63	0.43	0.65	0.24
XLE				1.00	0.55	0.63	0.46	0.48	0.42	0.33	0.43	0.13
XLF					1.00	0.84	0.63	0.75	0.59	0.34	0.62	−0.03
XLI						1.00	0.76	0.83	0.69	0.45	0.66	0.06
XLK							1.00	0.83	0.55	0.40	0.58	0.04
XLY								1.00	0.59	0.36	0.61	0.03
XLP									1.00	0.60	0.68	0.13
XLU										1.00	0.46	0.23
XLV											1.00	0.10
XAU												1.00

Notes: Pearson correlations of monthly log returns. Several features of this matrix motivate the satellite selection logic. XAU (gold) has near-zero correlation with all sector ETFs and a mildly negative correlation with bonds, confirming its role as a diversifier orthogonal to both core assets. XLE (energy) has the lowest correlation with the S&P 500 among cyclical sectors (0.63), explaining its frequent selection during the 2004–2009 commodity supercycle. XLU (utilities) has the lowest sector-to-S&P 500 correlation (0.51) among the cyclicals, consistent with its defensive classification. Bond–equity correlation is −0.06 unconditionally but varies substantially across regimes, as documented in **Table 4**.

4. Model

There are several criteria that predate the use of a HMM and are specific to this study. The outputs of the model need to have an intuitive economic interpretation that can be communicated to a portfolio manager, where we assume the portfolio manager is familiar with the concepts of expected return, standard deviation, and correlation, and understands their interpretation in a portfolio context. The identified regimes need to be stable across seeds and sample periods in order to reliably recognise key global events such as the dotcom crash, the Global Financial Crisis, and the COVID-19 pandemic. The outputs need to serve as input signals to the allocation layer. And the covariance between assets needs to be conditional on the regime, in order to capture the co-movement dynamics that inform the satellite tilt selection.

An important design point concerns what variables are used to define the regimes. This model is entirely endogenous: the regimes are identified exclusively from the joint distribution of the two return series, with no dividend yield, short-term interest rates, or macroeconomic variables introduced as exogenous predictors. This stands in contrast to a number of papers in the regime-switching literature, including the VAR-extended specification of Guidolin and Timmermann (2007) and the macro-conditioned models in the time-varying transition probability literature, which introduce external conditioning variables to capture predictability beyond the return distribution itself. The choice of an endogenous specification here is deliberate: it keeps the model self-contained within the information available from traded prices alone, which is consistent with the operational context of a wealth management firm that may not have reliable real-time macro feeds at the frequency and latency required for monthly rebalancing.

Regime-switching models for financial time series belong to a broader family of models that include threshold autoregressive models, hidden Markov models, hidden semi-Markov models, and smooth-transition models. Within this family, the Hidden Markov Model with Gaussian emissions was selected because it satisfies all four criteria above and because its filtered probability output is directly usable as a forward-looking allocation signal without additional transformation. GARCH-family models are widely used in the literature on co-movement and conditional heteroskedasticity, and this family of models produces valuable insights into the time-varying second moment structure of returns. However, GARCH outputs do not naturally produce discrete regime labels that partition the joint return distribution into economically interpretable states, which is a specific requirement of the allocation layer in this study. The HMM's filtered probabilities, by contrast, provide a direct probability distribution over a small number of interpretable states at each point in time.

4.1 HMM Specification

[PV: you marked this subsection and Estimation for removal in the competition version. Agreed — both are standard and can be compressed to a short paragraph plus a citation to Hamilton (1989), with the full specification moved to an appendix. Leaving intact for now since volume is not yet the constraint.]

Let $\mathbf{Y}_t = (y_{1t}, y_{2t})'$ denote the bivariate vector of monthly excess log returns on the S&P 500 Total Return index and LT09TRUU at time $t = 1, \dots, T$. The Hidden Markov Model assumes that the conditional distribution of $\mathbf{Y}_t$ is governed by an unobservable discrete state variable S_t that takes values in $\{0, 1, \dots, K-1\}$ and follows a first-order Markov chain with constant transition probability matrix $\mathbf{P}$, where element $p_{ij} = \Pr(S_{t+1} = j \mid S_t = i)$.

Conditional on the state, returns are assumed to follow a multivariate normal distribution:

$$\mathbf{Y}_t \mid S_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (1)$$

where $\boldsymbol{\mu}_k$ is the (2×1) vector of state-conditional mean excess returns and $\boldsymbol{\Sigma}_k$ is the (2×2) full covariance matrix for state k . The full covariance specification is essential: it allows the model to capture both state-specific volatility and state-specific cross-asset correlations, which are the mechanism through which the model captures regime-dependent bond-equity co-movement. As shown in **Table 4**, this correlation varies substantially across regimes and does not follow a simple negative-in-bear pattern.

The initial state distribution is parameterised as $\boldsymbol{\pi}_0 = (\pi_{0,0}, \dots, \pi_{0,K-1})'$ and estimated alongside the other parameters. The full parameter vector is $\boldsymbol{\theta} = \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}, \boldsymbol{\pi}_0\}_{k=0}^{K-1}$.

4.2 Estimation

The model is estimated by maximum likelihood via the Expectation-Maximisation algorithm introduced by Hamilton (1989). The implementation uses the `hmmlearn` Python library, with custom pre-processing for excess log return construction and post-processing for regime ordering and rolling window management. The EM algorithm iterates between two steps. The E-step computes the filtered state probabilities using the Hamilton filter:

$$\xi_{t|t}(k) = \Pr(S_t = k \mid \mathbf{Y}_1, \dots, \mathbf{Y}_t; \boldsymbol{\theta}) \quad (2)$$

These filtered probabilities are the forward-pass output of the filter: they represent the probability of being in regime k given all information available up to and including time t . They are computed recursively from the prediction step:

$$\xi_{t|t-1}(k) = \sum_j \xi_{t-1|t-1}(j) p_{jk} \quad (3)$$

updated with the likelihood of the current observation in each state.

The M-step updates θ by maximising the expected complete-data log-likelihood with respect to the smoothed state probabilities. Convergence is declared when the improvement in log-likelihood between iterations falls below 10^{-6} . To mitigate the risk of converging to local optima, 25 random initialisations were run and the solution with the highest converged log-likelihood was retained.

Likelihood computation via the forward algorithm. The observed-data log-likelihood is obtained by integrating out the latent state sequence. For a single observation $\mathbf{Y}_t$ in state k , the state-conditional emission density is:

$$f(\mathbf{Y}_t | S_t = k; \theta) = \frac{1}{(2\pi)^{n/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{Y}_t - \mu_k)' \Sigma_k^{-1} (\mathbf{Y}_t - \mu_k)\right) \quad (4)$$

where $n = 2$ is the number of observed series. The full observed-data log-likelihood $\mathcal{L}(\theta) = \log p(\mathbf{Y}_1, \dots, \mathbf{Y}_T; \theta)$ is evaluated using the forward (filter) recursion. Define the forward variable:

$$\alpha_t(k) = p(\mathbf{Y}_1, \dots, \mathbf{Y}_t, S_t = k; \theta) \quad (5)$$

The recursion is initialised at $t = 1$ as:

$$\alpha_1(k) = \pi_{0,k} \cdot f(\mathbf{Y}_1 | S_1 = k; \theta) \quad (6)$$

and propagated forward for $t = 2, \dots, T$:

$$\alpha_t(k) = f(\mathbf{Y}_t | S_t = k; \theta) \sum_{j=0}^{K-1} \alpha_{t-1}(j) p_{jk} \quad (7)$$

The log-likelihood is then:

$$\mathcal{L}(\theta) = \sum_{t=1}^T \log \sum_{k=0}^{K-1} \alpha_t(k) \quad (8)$$

In practice, the recursion is implemented in log-space using the log-sum-exp trick to avoid numerical underflow, which is the approach taken in `hmmlearn`. The filtered probabilities from Equation (2) are recovered by normalising: $\xi_{t|t}(k) = \alpha_t(k) / \sum_j \alpha_t(j)$.

The following code illustrates the core estimation call. `GaussianHMM` from `hmmlearn` is used with `covariance_type="full"` to allow a distinct full covariance matrix per regime. Twenty-five random initialisations are run and the best log-likelihood solution is retained.

```
from hmmlearn import hmm
import numpy as np

best_model, best_ll = None, -np.inf
```

```

for seed in range(1, 26):
    model = hmm.GaussianHMM(
        n_components=3,
        covariance_type="full",
        n_iter=2000,
        tol=1e-6,
        random_state=seed,
    )
    model.fit(X)          # X: (T, 2) array of excess log returns
    ll = model.score(X)
    if ll > best_ll:
        best_ll, best_model = ll, model

# One-step-ahead predictive probabilities
filtered    = best_model.predict_proba(X)
predictive  = filtered @ best_model.transmat_

```

Post-estimation, regimes are reordered by ascending mean equity return so that Regime 0 is always the bear state, consistently across all seeds and rolling windows.

4.3 Model Selection and the Two-Stage Design Rationale

Three state configurations were evaluated: $K = 2$, $K = 3$, and $K = 5$. The two-state model collapsed the bear and transitional periods into a single high-volatility state, losing the distinction between genuine market stress and regime uncertainty. The five-state model produced average regime durations of one to two months with no stable economic interpretation, consistent with the overfitting behaviour documented in Guidolin and Timmermann (2007). The three-state model, $K = 3$, produced economically interpretable and statistically stable regimes across the full sample.

It is important to explain the two-stage nature of this design, as a potential methodological concern arises from the fact that $k = 3$ was selected by examining the full 27-year sample, while the actual allocation strategy uses a rolling 60-month estimation window. These are two separate and complementary uses of the model. The full-sample estimation is a diagnostic exercise: its purpose is to establish that three states are the appropriate structural assumption for this asset pair, that these states are stable enough to carry economic interpretation, and that they recur across multiple market cycles in a consistent way. This is the model selection stage. Once $k = 3$ is established as the correct structural assumption, the rolling window then takes this as given and re-estimates only the parameter values, not the number of states, within each 60-month window.

Table 3: Model Selection: Log-Likelihood, AIC and BIC for $K = 2, 3, 5$ States

States	Log-lik.	Params	AIC	BIC	Avg. dur. (mo)	Min. regime frac.
$K = 2$	1427.86	13	-2829.72	-2780.53	11.0	0.234
$K = 3$	1445.83	23	-2845.67	-2758.64	15.3	0.062
$K = 5$	1466.68	49	-2835.35	-2649.94	4.7	0.009

Notes: Best log-likelihood across 25 random seeds for each K . Free parameters: $(K^2 - K)$ transition probabilities $+K \times 2$ means $+K \times 3$ unique covariance entries $+(K - 1)$ initial state probabilities. $AIC = -2\mathcal{L} + 2p$; $BIC = -2\mathcal{L} + p \ln T$, $T = 325$. $K = 3$ is selected: it has the best AIC (-2845.67) and adequate minimum regime fraction (6.2%). $K = 2$ has the best BIC but collapses the bear and transitional states into one, losing economic interpretability. $K = 5$ achieves the highest log-likelihood but produces a degenerate regime with only 0.9% of observations and an average duration of 4.7 months, consistent with overfitting. Bold = selected specification.

Models where any regime's expected duration was less than three to four months, or where any regime accounted for less than 6% of total observations, were considered too noisy and not developed further. The expected duration of regime k is:

$$\mathbb{E}[\text{duration}_k] = \frac{1}{1 - p_{kk}} \quad (9)$$

where p_{kk} is the diagonal element of the transition matrix for regime k . The three identified regimes are ordered by ascending mean excess equity return: Regime 0 is the bear state, Regime 1 is a transitional state, and Regime 2 is the bull state. This ordering is enforced consistently across all seeds and rolling windows.

Figure 2 shows the one-step-ahead predictive regime probabilities over the full 2004–2026 backtest period. The bear regime (Regime 0, red) spikes sharply during the GFC, COVID-19, and the 2022 rate shock, consistent with the rolling window adapting toward elevated bear-regime probability during these episodes. The transitional regime (Regime 1, orange) dominates the 2010–2016 recovery period, consistent with the sub-period alpha analysis in Section 8.

4.4 Model Output and Economic Interpretation

Table 4 reports the full regime-conditional moment structure, and **Table 5** presents the estimated transition probability matrix. The primary output of the HMM for allocation purposes is the one-step-ahead predictive probability vector:

$$\pi_{t+1|t} = \xi_{t|t} \cdot \mathbf{P} \quad (10)$$

This forward-looking vector encodes the model's current belief about which regime will prevail in the next period. It is constructed using only information available at time t , ensuring no look-ahead bias in the backtest.

Additional outputs extracted from the model include the regime-conditional moments (mean,

variance, skewness, excess kurtosis) for each asset, the correlation matrix Σ_k for each state, and the transition probability matrix $\mathbf{P}$.

Table 4: Summary Statistics for Monthly Excess Log Returns by HMM Regime (Model A)

Regime	Asset	Mean	Std Dev	Skew.	Exc. Kurt.	Min	Max	Corr.
<i>Return figures in % per month; Corr. = conditional Pearson correlation between S&P 500 TR and LT Bond.</i>								
Regime 0 — Bear <i>N = 20 (6.2%)</i>	S&P 500 TR	−2.916	7.949	0.115	−0.972	−18.466	9.130	0.28
	LT Bond	−0.523	3.443	0.772	−0.174	−4.993	7.657	
Regime 1 — Transitional <i>N = 141 (43.4%)</i>	S&P 500 TR	0.101	5.038	−0.191	−0.385	−13.303	12.062	−0.52
	LT Bond	0.573	1.647	0.027	−0.438	−3.519	4.747	
Regime 2 — Bull <i>N = 164 (50.5%)</i>	S&P 500 TR	1.301	2.612	−0.236	−0.100	−6.139	8.300	0.20
	LT Bond	−0.116	1.686	−0.415	0.765	−5.915	4.001	
Full Sample <i>N = 325 (100%)</i>	S&P 500 TR	0.521	4.384	−0.661	1.080	−18.466	12.062	−0.12
	LT Bond	0.158	1.854	−0.044	0.861	−5.915	7.657	

Notes: Excess log returns = $\log(1 + r_t) - \log(1 + r_{f,t})$; $r_{f,t}$ = one-month T-bill rate. Regime assignments: Viterbi hard-state sequence, best-seed HMM (Model A, seed = 24, without gold). Skewness and excess kurtosis follow Fisher’s definition. S&P 500 TR = $\sim$ SP500TR; LT Bond = LT09TRUU Bloomberg US Treasury 7–10 Year index. Positive correlation in Regime 0 (+0.28) reflects the 2022 rate shock: equities and bonds fell simultaneously. Sample: January 1999 – January 2026 ($T = 325$ monthly observations).

Table 5: HMM Transition Probability Matrix and Regime Persistence (Model A, 1999–2026)

From \ To	Regime 0 (Bear)	Regime 1 (Trans.)	Regime 2 (Bull)	Exp. Duration
Regime 0 (Bear)	0.892	0.000	0.108	9.2 months
Regime 1 (Transitional)	0.017	0.940	0.043	16.6 months
Regime 2 (Bull)	0.000	0.050	0.950	19.9 months

Notes: Estimated on the full 1999–2026 sample, best seed (seed = 24, log-likelihood = 1445.79, $T = 325$). Each row sums to 1; entries below 0.0005 are shown as 0.000. Expected duration = $1/(1 - p_{kk})$. Regime 2 (Bull) is the most persistent state at 19.9 months. The zero entry p_{02} (Bear $\rightarrow$ Bull) implies the model never transitions directly from a crisis state to a bull state without passing through the transitional regime.

The bear state (Regime 0) is characterised by negative mean excess equity returns and elevated volatility. Contrary to the textbook flight-to-quality narrative, the bond-equity correlation in Regime 0 is *positive* (+0.28), not negative. This reflects the composition of bear-state observations: the 20 months classified as Regime 0 include the 2022 rate shock, in which both equities and bonds fell simultaneously. The bear regime therefore conflates two types of stress — deflationary crashes where Treasuries hedge, and inflation/rate shocks where they do not.

Regime 1, the transitional state, is where the strongest negative bond-equity correlation appears (−0.52), providing genuine diversification between the two core assets. This is the regime covering 43% of the sample where the 60/40 benchmark most reliably delivers its intended diversification benefit. Regime 2, the bull state, has positive mean excess equity returns, lower volatility, and a mild positive correlation (+0.20), consistent with risk-on periods where both assets can be bid simultaneously.

Figure 2 — One-Step-Ahead Predictive Regime Probabilities (3-State HMM, Rolling 60-Month Window, 2004–2026)

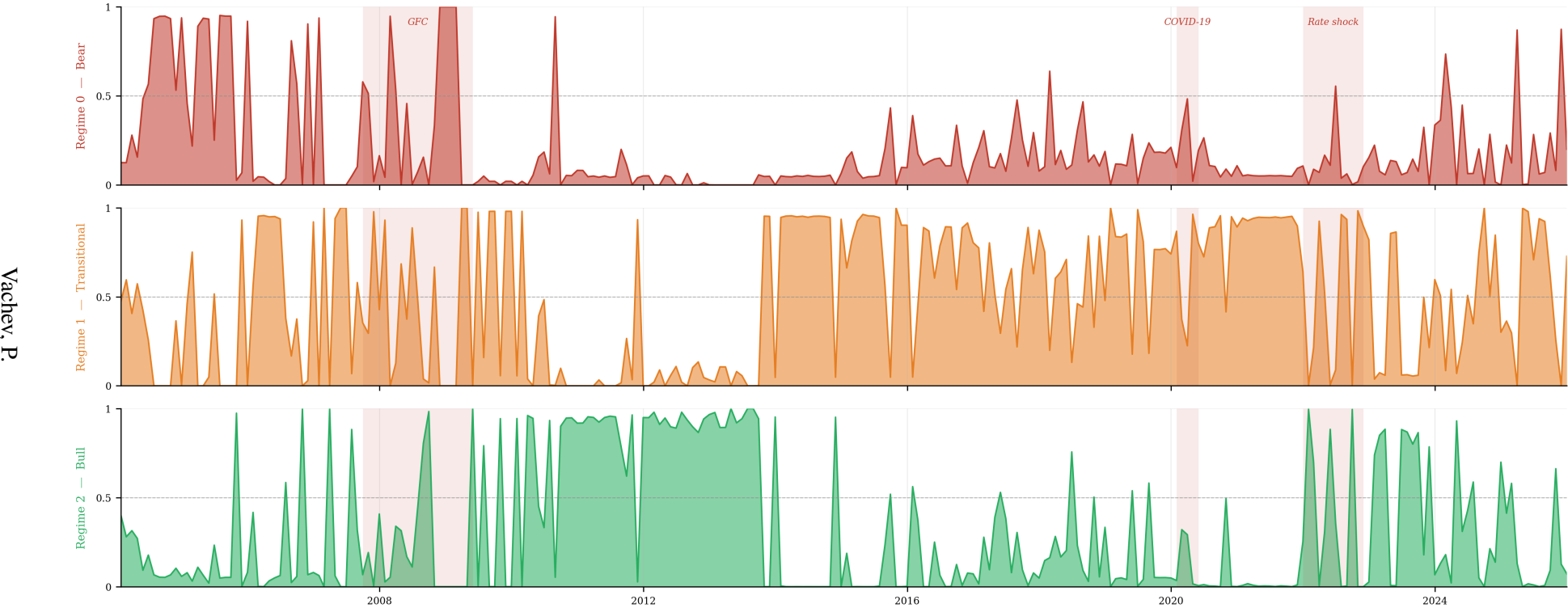


Figure 2: One-step-ahead predictive regime probabilities from the three-state HMM (rolling 60-month window, 2004–2026). Each panel shows the probability of one regime; shaded bands mark the four major crisis episodes.

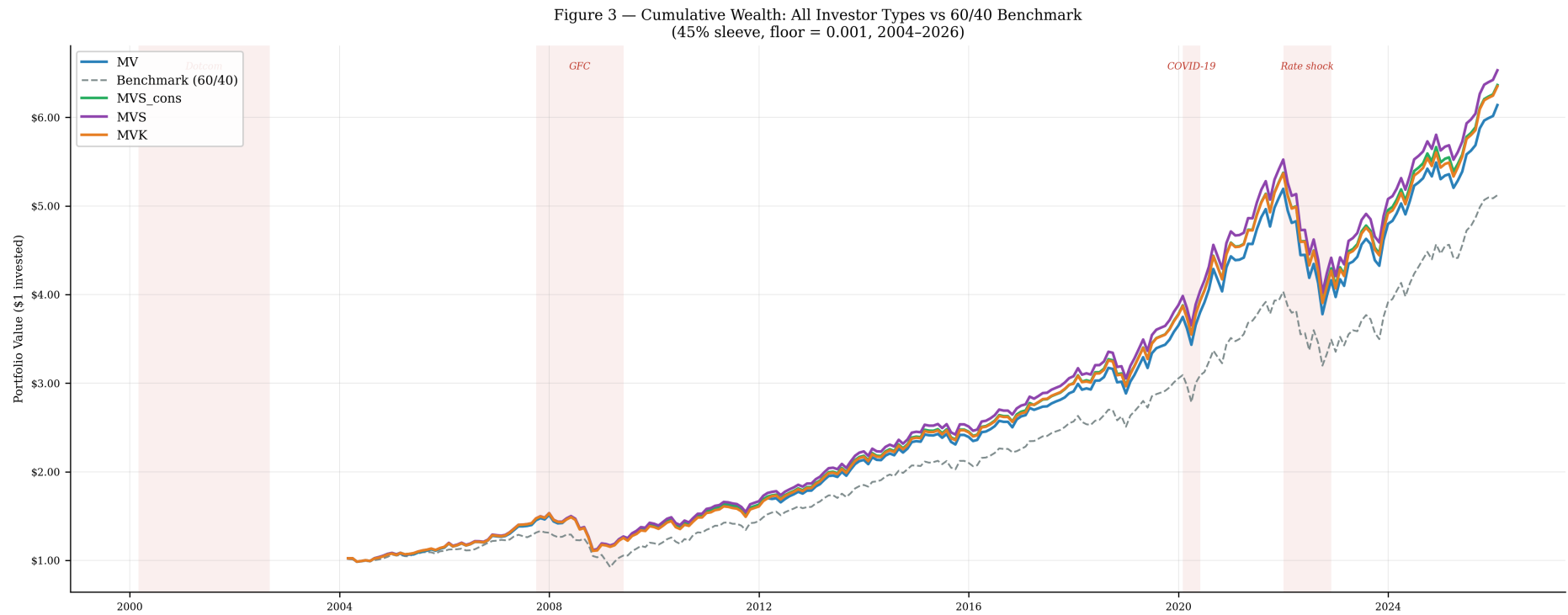


Figure 3: Cumulative wealth for all four investor types versus the 60/40 benchmark, 45% satellite sleeve, floor=0.001 (February 2004 – January 2026, \$1 invested). Shaded regions mark crisis episodes.

Figure 4 — Monthly Satellite Allocation Weights by Sector (MVS_cons, 45% sleeve, floor=0.001, 2004–2026)
 Bold frames: XLE (energy cycle 2004–09) · XLK (tech cycle 2016–26) · XAU (gold, crisis periods)

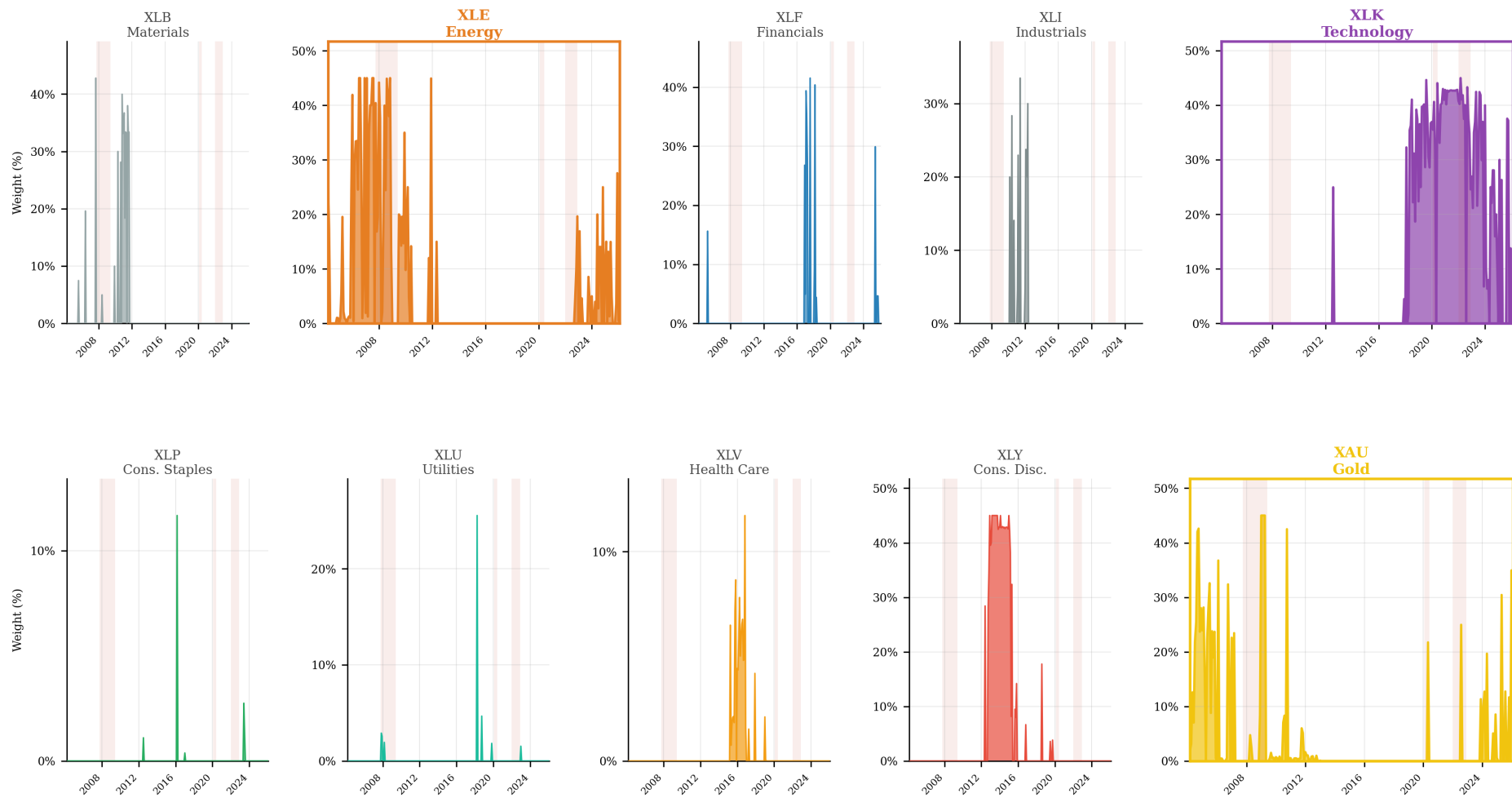


Figure 4: Monthly satellite allocation weights by sector (MVS_cons, 45% sleeve, floor=0.001, 2004–2026). Bold frames: XLE (energy cycle 2004–09), XLK (technology cycle 2016–26), XAU (gold, crisis periods).

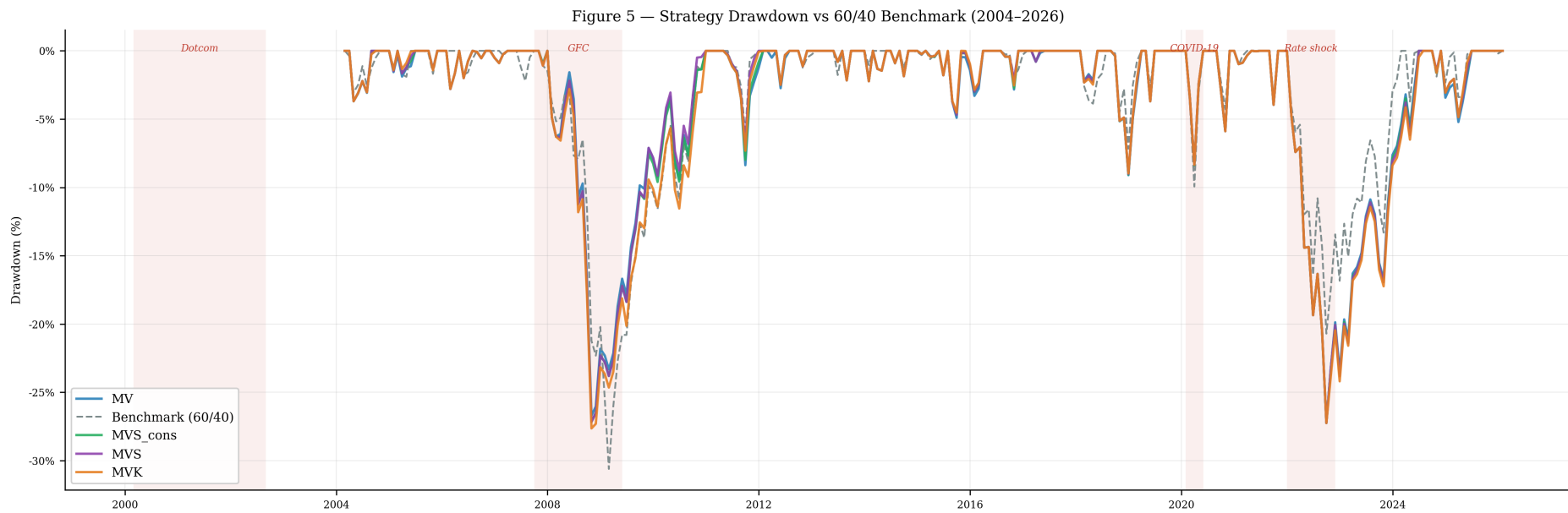


Figure 5: Strategy drawdown vs 60/40 benchmark (45% sleeve, floor = 0.001, 2004–2026). All four investor types shown.

5. Investor Preference Theory

This study considers several different types of investors, building from the mean-variance investor of Markowitz’s 1952 paper and then extending to higher moments. In this section we describe in what ways our approach differentiates from the relevant literature with respect to investor preference, and show that while the Taylor series expansion for higher moments within a CRRA framework works well for papers that use optimisation for weight allocation, our approach uses the same mathematical structure but calibrates it differently because we are scoring candidate portfolios rather than solving for optimal weights. The main reference for classifying investor preference is Scott and Horvath (1980). To quantify and calibrate investor preference, we turn to Jondeau and Rockinger (2006).

5.1 Formal Framework: Optimisation, Welfare, Certainty Equivalent, and Scoring

This subsection establishes the formal relationships among four concepts that are central to this thesis but are sometimes conflated in applied work: portfolio optimisation, investor welfare, certainty equivalent wealth, and the scoring criterion used here. The distinctions matter because this thesis deliberately departs from the optimisation framework adopted by most of the theoretical literature.

Portfolio optimisation. The classical formulation is a constrained optimisation problem over portfolio weights $\mathbf{w} \in \mathbb{R}^N$:

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \mathbb{E}[U(W_T)] \quad \text{subject to} \quad \mathbf{w}'\mathbf{1} = 1, \quad \mathbf{w} \geq \mathbf{0} \quad (11)$$

where $W_T = W_0(1 + R_p)$ is terminal wealth, $R_p = \mathbf{w}'\mathbf{r}$ is the portfolio return, and $U(\cdot)$ is a von Neumann-Morgenstern utility function. In a regime-switching setting, $\mathbb{E}[U(W_T)]$ is evaluated using the predictive distribution of returns, which depends on the current regime probabilities. Solving Equation (11) directly requires numerical optimisation at every rebalance date, which is computationally demanding and sensitive to moment estimation error — particularly for higher-moment utility functions where the objective surface is non-convex.

Investor welfare. The welfare of an allocation $\mathbf{w}$ is the expected utility of terminal wealth under that allocation:

$$V(\mathbf{w}) = \mathbb{E}[U(W_T(\mathbf{w}))] = \sum_{k=0}^{K-1} \pi_k \int U(W_0(1+r)) f(r|k) dr \quad (12)$$

where π_k is the predictive probability of regime k and $f(r|k)$ is the regime-conditional return density. Unlike optimisation, welfare evaluation takes $\mathbf{w}$ as given and returns a scalar measure of investor wellbeing. Guidolin and Timmermann (2007, 2008) compute welfare costs by comparing $V(\mathbf{w}_{\text{regime}}^*)$ against $V(\mathbf{w}_{\text{static}}^*)$ for the optimal portfolio under each specification.

Certainty equivalent. The certainty equivalent $CE(\mathbf{w})$ converts welfare into a money-metric by finding the risk-free return r^* such that the investor is indifferent between the risky portfolio and a guaranteed payoff:

$$U(W_0(1 + r^*)) = V(\mathbf{w}) \implies CE(\mathbf{w}) = U^{-1}(V(\mathbf{w})) / W_0 - 1 \quad (13)$$

Welfare cost comparisons expressed in “cents per dollar” in the literature (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007) are certainty equivalent differences: $\Delta CE = CE(\mathbf{w}_{\text{regime}}^*) - CE(\mathbf{w}_{\text{static}}^*)$.

Taylor expansion and the scoring criterion. Rather than evaluating Equation (12) by numerical integration, this thesis adopts the Taylor series approximation of Jondeau and Rockinger (2006). For a twice continuously differentiable utility function U , expanding $\mathbb{E}[U(R_p)]$ around μ_p to the fourth order gives:

$$\mathbb{E}[U(R_p)] \approx U(\mu_p) + \frac{U''(\mu_p)}{2} \sigma_p^2 + \frac{U'''(\mu_p)}{6} \mu_3(R_p) + \frac{U^{(4)}(\mu_p)}{24} \mu_4(R_p) \quad (14)$$

where μ_3 and μ_4 are the third and fourth central moments of R_p . Normalising by $U'(\mu_p)$ and absorbing the utility derivatives into preference parameters λ, γ, δ yields Equation (15).

The key departure from the optimisation literature is that this thesis uses Equation (15) as a *scoring criterion* rather than an objective function to be maximised. At each rebalance date, a finite set of candidate portfolios (defined by the discrete satellite weight grid) is evaluated by computing their predictive moments and ranking them by score. The portfolio with the highest score is selected, provided it clears the improvement floor. This is computationally equivalent to evaluating $V(\mathbf{w})$ for each candidate and selecting the maximiser, but without solving the continuous optimisation problem in Equation (11). The approach is exact within the grid of candidates and approximate relative to the global optimum, with the approximation error shrinking as the grid becomes finer.

5.2 Direction of Higher-Moment Preference

Scott and Horvath published a note in 1980 titled *On the Direction of Preference for Moments of Higher Order than the Variance*, in which they explore the third and fourth moments of the utility curve within the mean-variance framework. Using proof by contradiction within an expected utility framework and under assumptions of risk aversion, they determine the sign of preference for higher moments *a priori*, without requiring any specific parametric form for the utility function.

By showing that the third moment of the utility function is positive, the authors establish that a mean-variance-skewness investor would, under some circumstances, trade lower expected return or higher variance in exchange for positive skewness in the return distribution. Such an

investor would tolerate a distribution that produces more frequent small losses in exchange for a small probability of large gains. In the context of this study, this corresponds to an investor who is relatively growth-oriented while remaining passive, and who would accept asymmetric return distributions more readily than a kurtosis-averse investor.

The fourth moment result is particularly relevant to this study. Scott and Horvath show that the fourth moment of the utility of wealth has a negative sign, meaning that investors dislike positive excess kurtosis. Kurtosis measures tail events symmetrically, both positive and negative. The negative sign result is important because the target beneficiaries of this study are passive clients in a wealth management context. From a practical perspective, positive tail events might signal low portfolio diversification, while negative tail events cause large material losses and damage the client-manager relationship. A kurtosis-averse investor, when implemented through the scoring function, should systematically avoid satellite tilts whose kurtosis contribution is high, providing a form of tail risk management that does not require explicit CVaR or downside optimisation.

5.3 Scoring Under Higher-Moment Preferences

Jondeau and Rockinger (2006) derive a Taylor series expansion of expected utility around the moments of the portfolio return distribution, providing a closed-form scoring function that is accurate even for moderately non-normal distributions. Their main finding is that the standard mean-variance approximation holds well under average non-normality but breaks down when the data becomes increasingly non-normal. Their polynomial utility approach handles a large number of assets while remaining computationally manageable, which is an important operational advantage for this study.

The Taylor expansion of expected utility around the portfolio return distribution, truncated at the fourth order, gives:

$$\mathbb{E}[U(R_p)] \approx \mu_p - \lambda \sigma_p^2 + \frac{\gamma}{6} \text{sk}_p - \frac{\delta}{24} (\kappa_p - 3) \quad (15)$$

where μ_p is the expected portfolio return, σ_p^2 is the variance, sk_p is the skewness, and $(\kappa_p - 3)$ is the excess kurtosis. The factorial denominators $\frac{1}{6}$ and $\frac{1}{24}$ are a direct result of the Taylor expansion and are critical for correct calibration. Without them, the γ and δ coefficients would be 6 and 24 times too large relative to λ , causing the higher-moment terms to dominate the score entirely.

Three investor types are defined as nested truncations of Equation (15). The MV investor truncates at the second order, ignoring higher moments entirely. The MVS investor adds the skewness term. The MVK investor adds both the skewness and kurtosis terms. A conservative MVS variant, MVS_cons, uses a lower γ coefficient calibrated to represent 15% of the variance term rather than 20%, representing a more moderate skewness preference. This

study deliberately separates MVS and MVK investor types, for two reasons: first, to isolate which moment is driving any observed improvement in risk-adjusted performance; and second, because the signs of skewness and kurtosis preferences impose conflicting demands on the return distribution in some configurations, and keeping them separate allows clean attribution.

An important interpretive caveat: the scoring function is *not* a structural estimate of CRRA or any parametric utility function. It is a normalised, utility-inspired ranking criterion where γ and δ are calibrated to keep higher-moment terms at 15–20% of the variance penalty at benchmark moments (**Table 6**). This is moment-contribution targeting, not utility-derivative calibration in the Jondeau and Rockinger (2006) sense.

5.4 Calibration of Preference Parameters

The choice of values for the preference parameters was grounded in the existing literature. Jondeau and Rockinger (2006) use λ values of 1, 3, and 5, where 3 is consistent with moderate risk aversion. For the higher-moment coefficients, the approach was to scale γ and δ such that the skewness and kurtosis terms represent a meaningful but secondary contribution to the score, consistent with the Taylor expansion’s role as a correction to the mean-variance baseline rather than a replacement for it.

The calibrated parameter values are reported in **Table 6**. Specifically, at the unconditional moments of the benchmark 60/40 portfolio over the full sample period ($\lambda = 3.0$, $\sigma^2 = 0.000672$, skewness = -0.70 , excess kurtosis = 1.95 , $\mu = 0.006543$), the variance penalty represents approximately 31% of the mean return contribution. We calibrate γ and δ so that the higher-moment terms each represent approximately 20% of the variance term, or roughly 6% of the mean.

Table 6: Score Term Calibration at Benchmark Unconditional Moments

Score term	Formula	Value	% of mean	% of variance
Mean return	μ_p	0.006543	100% — primary	—
Variance penalty	$\lambda \cdot \sigma^2$	0.002016	31%	100% (anchor)
Skewness term	$\gamma \cdot \text{sk} $	0.000403	6%	20% (mod. γ)
Kurtosis term	$\delta \cdot \kappa_e $	0.000403	6%	20% (mod. δ)

Note: Benchmark moments: $\lambda = 3$, $\sigma^2 = 0.000672$, skew = -0.70 , excess kurtosis = 1.95 , $\mu = 0.006543$. Calibrated parameters: $\gamma_{\text{cons}} = 0.000431$ (15%), $\gamma_{\text{mod}} = 0.000574$ (20%), $\delta_{\text{mod}} = 0.000207$ (20%).

6. Allocation Layer

The allocation architecture follows a core-plus-satellite structure designed to be operationally compatible with the standardised portfolio context described in the introduction. The core is a fixed 60% allocation to the S&P 500 Total Return index and 40% allocation to LT09TRUU. These weights are institutional constants. The bond allocation is fixed throughout the entire backtest, which cleanly separates the regime signal, which operates within the equity sleeve, from duration management, which is a separate risk decision.

The satellite sleeve introduces up to two sector or commodity tilts that displace only the S&P 500 component of the core. The primary configuration uses a maximum satellite sleeve of 45%, so the S&P 500 allocation ranges from 60% when no tilt is active to a minimum of 15% when the full sleeve is deployed. LT09TRUU remains at 40% in all configurations. No leverage is introduced at any point.

6.1 Filtered and Predictive Probabilities

At each monthly rebalance date, the HMM is refitted on the most recent 60 months of excess log returns, producing updated parameter estimates and a filtered probability vector for the current month. As described in Hamilton (1989), the filtered probabilities $\Pr(s_t = i \mid \mathcal{F}_t)$ represent the probability of being in regime i given all information available up to and including time t . These are the Hamilton filter outputs from the E-step of the EM algorithm.

A key aspect of the approach, following Ma et al. (2011), is that the model does not directly predict returns. Instead, it estimates predictive probabilities: one-step-ahead regime probabilities computed by propagating the filtered probabilities through the transition matrix:

$$\pi_{t+1|t} = \xi_{t|t} \cdot \mathbf{P} \quad (16)$$

These predictive probabilities are the primary input to the scoring engine. They represent the model's best forward-looking estimate of regime probabilities for the next period, constructed using only information available at the current rebalance date.

6.2 Candidate Enumeration and Scoring

The scoring engine enumerates all feasible candidate portfolios. For each sensitivity configuration, the total satellite sleeve cap $S \in \{20\%, 25\%, 30\%, 35\%, 40\%, 45\%, 50\%\}$ defines the maximum combined satellite allocation. Individual satellite weights are drawn from a discrete grid of $\{0\%, 5\%, 10\%, \dots, S\}$ in 5% increments, with up to two satellites selected simultaneously subject to the total cap. A single satellite may reach the full sleeve cap S if no second satellite is selected. This produces a finite, manageable candidate set at each rebalance date. The primary configuration uses $S = 45\%$; the conservative configuration uses $S = 20\%$. Each candidate is evaluated by computing its regime-conditional portfolio return

moments and applying the scoring function for the relevant investor type.

For each candidate portfolio with weight vector $\mathbf{w}$, the predictive moments are computed by aggregating regime-conditional moments across states using the predictive probability weights:

$$M_k = \sum_j \pi_j \cdot \mathbb{E} \left[R_p^k \mid \text{state } j \right] \quad (17)$$

where M_k is the k -th aggregated raw moment and π_j is the predictive probability of state j . Variance, skewness, and excess kurtosis are then derived from the aggregated raw moments M_1 through M_4 using the standard central moment relationships:

$$\sigma_p^2 = M_2 - M_1^2 \quad (18)$$

$$\mu_3 = M_3 - 3M_1M_2 + 2M_1^3 \quad (19)$$

$$\mu_4 = M_4 - 4M_1M_3 + 6M_1^2M_2 - 3M_1^4 \quad (20)$$

$$\text{sk}_p = \mu_3 / (\sigma_p^2)^{3/2} \quad (21)$$

$$\kappa_{e,p} = \mu_4 / (\sigma_p^2)^2 - 3 \quad (22)$$

This aggregation is applied to raw moments before deriving central moments, which is the correct approach for a Gaussian mixture distribution.

6.3 Score Improvement Floor

The candidate with the highest investor score is selected, subject to a score improvement floor. The floor requires that the improvement in expected utility score relative to the no-tilt baseline must exceed a minimum threshold. When no candidate clears the floor, the portfolio holds the passive 60/40 core for that period. This decision is recorded in the backtest as a NONE observation.

The score improvement floor functions analogously to a hurdle rate in active management: the model only deviates from the benchmark when the regime-conditional expected utility advantage of the satellite clearly justifies the operational cost and risk of the tilt. In the primary configuration, the floor is set at 0.001 in utility score units.

Transaction costs are applied at a rate of 5 basis points per side (buy and sell) on each portfolio rebalance. In the primary configuration (45% sleeve, floor = 0.001, MVS investor), average monthly portfolio turnover is approximately 29.3%. At 5 basis points per side, the monthly transaction cost is approximately $0.293 \times 2 \times 5 = 2.93$ basis points, or roughly 35 basis points per annum. This is consistent with institutional execution costs in liquid US equity ETFs and Treasury indices. All reported performance figures are net of these transaction costs.

7. Allocation Tilts and Computational Elements

7.1 Satellite Universe and Regime Styles

The ten candidate satellites are classified by their expected behaviour across the bear-bull regime dimension. Cyclical sectors — Energy (XLE), Technology (XLK), Financials (XLF), Industrials (XLI), Materials (XLB), and Consumer Discretionary (XLY) — are expected to outperform in bull regimes and underperform in bear regimes. Defensive sectors — Consumer Staples (XLP), Utilities (XLU), and Health Care (XLV) — are expected to preserve value in bear regimes and lag in bull regimes. Gold (XAU) is classified as a defensive satellite, reflecting its historically low and regime-varying correlation with the equity market.

The inclusion of gold in the satellite universe is supported by Baur and Lucey (2010), who establish that gold acts as both a hedge and a safe haven for equity investors under normal conditions. Echaust and Just (2022), using extreme value theory on 46 stock indices, find that gold reduced extreme downside risk during the GFC by shifting portfolio return tails from heavy Fréchet-type to thin Gumbel-type, but the same result did not hold during COVID-19, where portfolio tails remained heavy even after gold allocation. This crisis-specific conditionality directly motivates the regime-based treatment of gold in the present study. However, this classification is not unconditional. Kinatader, Gurrib, and Choudhury (2024), using the same SPDR sector ETF universe as the present study, find that gold provides safe-haven protection for financials, consumer discretionary, industrials, technology, and health care, but not for energy, materials, utilities, or consumer staples. This sector-specific conditionality directly motivates treating gold as a scored candidate rather than a permanent strategic allocation. Kinatader, Campbell, and Choudhury (2021), in a broader study covering cryptocurrencies alongside precious metals, provide additional evidence of crisis-specific variation in gold's tail-risk properties. In the model developed here, gold is treated as a conditional satellite: it is selected only when the regime signal and the scoring function jointly indicate that its inclusion improves the investor's expected utility.

7.2 Conviction Scalar

After the scoring engine selects the best candidate tilt, the applied satellite weight is not equal to the scored weight. A conviction scalar modulates the actual allocation as a function of regime certainty. For cyclical satellites, the applied weight is scaled by $(1 - p_{\text{bear}})$; for defensive satellites including gold, it is scaled by p_{bear} :

$$w_{\text{applied}}^{\text{cyclical}} = w_{\text{scored}} \cdot (1 - p_{\text{bear}}) \quad (23)$$

$$w_{\text{applied}}^{\text{defensive}} = w_{\text{scored}} \cdot p_{\text{bear}} \quad (24)$$

This mechanism ensures that the full scored weight is applied only when the model is highly confident about the prevailing regime. When regime probabilities are diffuse, the applied

satellite weight shrinks toward zero and the portfolio gravitates toward the passive 60/40 core.

7.3 Rolling Window Rationale and the Structural Break Problem

The rolling 60-month estimation window is the central design choice of this thesis. Under an expanding window, the HMM is estimated on all available history from the start of the sample at each rebalance date. XLE's extraordinary 2003–2008 performance permanently elevates its regime-conditional moments; the model continues to select XLE through the 2010–2015 period, generating active alpha of approximately -0.035% per month over a six-year stretch when energy significantly underperformed. A 60-month rolling window resolves the stale-moment problem by limiting the estimation to the five years most recent to each rebalance date.

The 60-month window length was selected as a balance between recency and statistical reliability. A shorter window of 24–36 months would provide faster adaptation but insufficient observations for stable higher-moment estimation within each regime. A window of 60 months is a compromise between adaptation and sample size. Because bear regimes represent approximately 6% of full-sample observations, a 60-month window contains only 3–4 bear-state months on average — insufficient for standalone higher-moment estimation in isolation. This is a genuine limitation and a reason the model uses the full predictive probability vector rather than hard regime classification: by weighting all three regime distributions by their predictive probabilities, the scoring function draws on all 60 observations rather than only the bear-state subset.

8. Key Findings

8.1 HMM Model Findings

An interesting finding from comparing different initial runs of the HMM is that under different large sample periods, the model produces different regime structures. A model starting in 1992 instead of 1999 shows a very specific post-COVID regime characterised by high correlations and elevated standard deviations across equities and bonds, with near-zero equity returns. This finding is documented here to emphasise that the rolling window design naturally avoids anchoring the regime structure to historical periods that are no longer representative of current market dynamics.

The three-state model fitted on the 1999–2026 sample produces three economically interpretable regimes. Regime 0 covers crisis periods including the dotcom crash of 2000–2002, the Global Financial Crisis of 2008–2009, the 2020 COVID-19 drawdown, and the 2022 rate shock. Regime 1 is a transitional state that appears around regime turning points. Regime 2 captures the sustained bull markets of 2003–2007, 2010–2019, and 2020–2021.

8.2 Primary Allocation Results

The primary configuration uses a 45% maximum satellite sleeve (our preferred configuration), a score improvement floor of 0.001, a rolling 60-month estimation window, and up to two simultaneous satellite tilts. *Note: the sleeve size, floor, and investor type were selected by inspecting a sensitivity grid of 48 configurations on the same historical path. Results should be interpreted as evidence of economic plausibility rather than clean out-of-sample performance. The full sensitivity table (Table 8) is presented transparently.* The backtest period runs from February 2004 to January 2026, a total of 264 monthly rebalance decisions. **Table 7** presents the performance summary across investor types and the main sensitivity variants.

Table 7: Performance Summary — Primary Configuration and Sensitivity Analysis

Config.	CAGR	vs BM	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
Benchmark (60/40)	7.71%	—	0.875	−30.61%	7.11%	1.104	0.252	—
45%/0.001 / MV	8.60%	+0.89%	0.920	−27.25%	7.78%	1.123	0.316	1.5%
45%/0.001 / MVS_cons	8.78%	+1.07%	0.939	−27.25%	7.77%	1.145	0.322	3.0%
45%/0.001 / MVS [†]	8.90%	+1.19%	0.956	−27.25%	7.72%	1.167	0.327	3.4%
45%/0.001 / MVK	8.76%	+1.05%	0.941	−27.64%	7.76%	1.145	0.317	3.0%
20%/0.001 / MV*	8.06%	+0.35%	0.923	−26.78%	7.05%	1.159	0.301	32.6%
20%/0.001 / MVS_cons*	8.09%	+0.38%	0.927	−27.00%	6.94%	1.181	0.300	30.7%
20%/0.001 / MVS*	8.10%	+0.39%	0.927	−27.00%	6.95%	1.181	0.300	30.7%
20%/0.001 / MVK*	8.04%	+0.33%	0.923	−27.38%	6.93%	1.176	0.294	29.2%

Notes: All configurations: rolling 60-month window, 60/40 core, 5 bp transaction costs, 2004–2026. [†] Best Sharpe ratio across all configurations (0.956). * Conservative 20% sleeve: the only group where all investor types simultaneously beat the benchmark on downside deviation. NONE% = proportion of months holding the passive 60/40 core.

All four investor types in the primary configuration beat the benchmark on CAGR, Sharpe ratio, maximum drawdown, Sortino ratio, and Calmar ratio. The 45% primary configuration accepts a modest increase in downside deviation (7.72–7.78% versus the benchmark 7.11%) in exchange for substantially higher CAGR (+89–119 basis points) and Sharpe ratio (+45–81 basis points). The 20% sleeve variant, reported in the lower panel of **Table 7**, is the only configuration where all four investor types simultaneously beat the benchmark on all five risk-adjusted metrics including downside deviation.

Figure 3 shows the cumulative wealth trajectories for all four investor types versus the 60/40 benchmark. The strategies track the benchmark closely through 2016 and then diverge materially in the post-2016 period as the rolling window locks onto XLK dominance.

8.3 Sub-Period Alpha Analysis

The outperformance is not uniformly distributed across time. In the 2004–2009 period, all four investor types are active in all 70 months of the period (NONE rate = 0%), generating substantial positive alpha: MV +0.232%/month, MVS_cons +0.243%, MVS +0.252%, and MVK +0.213%. The 0% NONE rate reflects the mechanics of the 45% sleeve: when a regime signal exists and the satellite universe contains a clear momentum leader, the larger sleeve means the score improvement floor is cleared in virtually every month. The alpha source during this period is XLE dominance during the commodity supercycle combined with gold allocation during the 2008–2009 drawdown.

The 2010–2015 period is the most instructive sub-period for evaluating the contribution of higher-moment scoring. All 71 months are active. MV records –0.035% per month and MVS_cons –0.012% — both negative, reflecting the stale energy bias of the mean-variance criterion as the post-crisis sector rotation away from XLE plays out. MVS and MVK both flip to positive territory at +0.013% and +0.014% respectively. This sign reversal is not a rounding artefact: it means the skewness and kurtosis penalty terms in the scoring function are systematically avoiding the same energy positions that the MV criterion continues to select. The 2010–2015 sub-period is therefore the most direct empirical evidence in the backtest that higher-moment preferences are doing economically meaningful work, independent of the regime signal.

In the 2016–2026 period (120 months), the model is inactive for only 4–9 months depending on investor type (NONE rate $\approx$ 3–7%). All four investor types generate positive alpha: MV +0.036%/month, MVS_cons +0.048%, MVS +0.050%, and MVK +0.049%. The rolling window adapts toward XLK exposure in the post-2016 period, consistent with the realised technology-led market cycle and re-engages gold allocation from 2022 onwards, with higher-moment investors maintaining a consistent edge over MV throughout.

Figure 4 shows the month-by-month satellite allocation weights for MVS_cons. The three-phase structure of sector leadership is immediately visible: XLE (energy, orange) dominates

2004–2009, nearly all sectors go quiet through 2010–2015, and XLK (technology, purple) becomes the dominant satellite from 2016 onwards. XAU (gold, yellow) appears persistently at low weights throughout and spikes during crisis episodes.

The sub-period pattern also illustrates why gold is an indispensable component of the satellite universe. Without gold, all configurations underperform the benchmark on a full-sample basis. The model earns its largest premiums precisely when gold and a sector ETF are selected simultaneously in bear or transitional regimes.

Figure 5 shows the rolling drawdown for all four investor types. The strategies achieve materially smaller peak drawdowns during the GFC (all types: -27.25% versus benchmark -30.61%) and recover faster, while the 2022 rate shock produces broadly similar drawdown profiles across strategies and benchmark.

8.4 Rolling Window versus Expanding Window and the A1 Frozen Model

The expanding window backtest fails in a specific and instructive way. Because all history from 1999 is used at each rebalance date, XLE's extraordinary 2003–2008 performance permanently elevates its regime-conditional moments, and the model persists in selecting XLE through 2015, generating approximately -0.035% monthly alpha drag during a six-year period of energy underperformance.

The A1 frozen-parameter backtest provides the sharpest contrast. The HMM and satellite moments are estimated on the 1999–2016 training window and frozen for out-of-sample application from 2017 onwards. The frozen model records a NONE rate of 93% in the 2017–2026 test period, selecting gold only eight times in 108 monthly decisions and otherwise holding the passive 60/40 core. Strikingly, all four investor types produce identical results in the A1 test: 9.06% CAGR, Sharpe 0.921. This identity confirms that the frozen model's near-complete disengagement is not a preference problem but a signal problem.

8.5 Higher-Moment Investor Differentiation

The higher-moment investor types produce measurably different outcomes from the MV baseline. The MVK investor achieves the lowest downside deviation across all primary configurations, recording 6.98% versus the benchmark's 7.11%, a 13 basis point improvement. As documented in Section 8, the 2010–2015 sub-period provides the sharpest evidence of this: MVS and MVK are the only investor types to record positive alpha in that period, while MV and MVS_cons remain negative.

8.6 Sensitivity Analysis

The sensitivity analysis across sleeve sizes and score improvement floors reveals a monotonic trade-off between performance and selectivity. As the sleeve size increases from 20% to 40% with floor = 0.001, CAGR improves from 8.06% to 8.57% for the MV investor, but

downside deviation worsens from 7.05% to 7.58%, exceeding the benchmark's 7.11% at every sleeve size above 20%. The NONE rate collapses from 33% to 3%. The 20% sleeve with floor = 0.001 is the only configuration that beats the benchmark on all five risk-adjusted metrics simultaneously, including downside deviation. Larger sleeves improve CAGR and Sharpe but exceed the benchmark downside deviation as the satellite allocations become more concentrated. The full sensitivity results across all 48 configurations are presented in **Table 8** in Appendix A. The monotonic pattern — increasing CAGR and Sharpe with sleeve size, deteriorating downside deviation — is visible across both floor levels.

Performance ratio definitions. The ratios used throughout this thesis are defined as follows:

$$\text{Sortino} = \frac{\bar{R}_p - r_f}{\sigma_{\text{down}}} \quad \text{where } \sigma_{\text{down}} = \sqrt{\frac{1}{T} \sum_t \min(R_{p,t} - r_f, 0)^2}$$

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{Max Drawdown}|}$$

$$\text{Sterling} = \frac{\text{CAGR}}{\text{top-3 drawdown episodes}}$$

$$\text{Burke} = \frac{\text{Sharpe} \times \sigma}{\sqrt{\sum_i (\text{top-3 drawdown depths})_i^2}}$$

$$\text{Omega} = \frac{\sum_t \max(R_{p,t} - \tau, 0)}{\sum_t \max(\tau - R_{p,t}, 0)}, \quad \tau = 0$$

All ratios are computed on the full monthly return series. The Sortino ratio penalises only downside deviation, making it the most relevant metric for the kurtosis-averse MVK investor. The Calmar ratio captures the return-per-unit-of-peak-loss trade-off central to the conservative 20% sleeve discussion.

9. Gold as a Conditional Safe Haven

An interesting finding of this research concerns the usefulness of gold as a safe haven asset. The results of several HMM runs, described in Section 8, begin to illustrate this point.

Baur and Lucey (2010) establish that gold is a hedge and a safe haven for stocks in the US and UK markets under normal conditions. Echaust and Just (2022) provide complementary evidence using extreme value theory: applying the block maxima method to 46 stock indices, they show that gold shifted portfolio return tails from Fréchet-type (heavy) to Gumbel-type (thin) during the GFC, but failed to do so during COVID-19, where tails remained heavy regardless of gold allocation. This asymmetry — gold as a genuine tail hedge in some crises but not others — is precisely what motivates the conditional, regime-dependent treatment of gold in the present study. Kinatader, Gurrib, and Choudhury (2024), using the same SPDR sector ETF universe, find that gold's safe-haven role is sector-specific: it protects financials, technology, and health care but not energy, materials, or utilities — a finding consistent with the sector-rotation dynamics documented in Section 8. Kinatader, Campbell, and Choudhury (2021) document crisis-specific variation in gold's tail-risk properties across a broader asset universe. The present study adds a regime-switching dimension to this debate.

The rolling window HMM identifies periods in which gold's correlation with the broad equity market deteriorates. In the 2013–2019 period, which corresponds to gold's structural bear market following the 2011 peak, the model assigns gold near-zero weight and the score improvement floor keeps the satellite inactive in most months. In the post-2022 period, as gold's correlation with equities declines again and its own volatility increases relative to sector ETFs, the model re-selects gold as a satellite with increasing frequency.

What the regime model reveals is that the question of whether gold is a safe haven cannot be answered in absolute terms. It is conditional on which regime the model is in, what gold's recent correlation with equities looks like within the rolling estimation window, and whether gold's regime-conditional moments score above the improvement floor relative to the passive core. The model's endogenous treatment of gold, as one candidate among ten in a competitive scoring environment, is arguably a more defensible allocation approach than treating it as a permanent strategic allocation or as a crisis-period override.

10. Conclusion

This thesis develops and empirically evaluated a regime-aware tactical allocation framework for the equity sleeve of a 60/40 balanced portfolio. The framework combines a three-state Hidden Markov Model for regime identification, a sector ETF and gold satellite universe, a Taylor-expanded expected utility scoring function accommodating three investor preference types, and a rolling 60-month estimation window enabling adaptation to structural breaks in sector leadership.

The primary empirical finding is that the rolling window is a necessary design choice, not merely a convenient one. The expanding window fails because it cannot forget the 2003–2008 commodity supercycle, and the frozen-parameter model disengages almost entirely from active allocation in the post-2016 period. Both failures are predictable consequences of applying stale moment estimates to a sector universe whose leadership structure changes across economic cycles. The rolling window addresses both failure modes while maintaining the epistemic honesty of using only information available at each rebalance date.

The secondary finding concerns the value of higher-moment investor types. MVS and MVK investors consistently outperform the MV baseline on downside-focused metrics, with MVK achieving the lowest downside deviation in every configuration tested. The most direct evidence comes from the 2010–2015 sub-period: MVS and MVK flip to positive alpha (+0.013% and +0.014% per month) while MV and MVS_cons remain negative (−0.035% and −0.012%), a sign reversal attributable to the skewness and kurtosis penalty terms systematically avoiding the stale energy positions that the mean-variance criterion continues to select. Across all three sub-periods, higher-moment investor types consistently match or exceed the MV baseline.

The framework has important limitations. The backtest covers a single historical path, and the configuration search across sleeve sizes and floors constitutes a form of in-sample optimisation on the same data used for evaluation. This is addressed by presenting the full sensitivity table transparently. A third limitation concerns benchmark fairness. The strategy is compared only to the static 60/40 benchmark. A static XLK sleeve would test whether post-2016 outperformance reflects technology exposure rather than dynamic selection; a static gold sleeve would test whether drawdown improvement reflects gold alone. These comparisons are not implemented here but remain important robustness checks. The sub-period decomposition provides partial evidence against a pure XLK story (positive alpha in 2004–2015 predates XLK dominance), but a formal comparison is left to future work.

A second limitation is the myopic one-period scoring function, which does not account for multi-period horizon effects or regime transition hedging demands as the Bellman equation approach of Guidolin and Timmermann (2007) does. This is a deliberate trade-off for operational manageability. It is worth noting that the EM algorithm used to estimate the HMM

is distinct from Bellman equation optimisation: the EM algorithm is a parameter estimation procedure for the latent-state model, whereas the Bellman approach refers to solving the dynamic programming problem for optimal portfolio weights across a multi-period horizon. The present study replaces the latter with a one-period scoring function.

Two directions for future work are identified. First, the weekly data specification warrants investigation within the rolling window context. A 60-month rolling window contains approximately 260 weekly observations, providing substantially more statistical power while limiting exposure to structural breaks. Second, a nonlinear conviction scalar could improve model behaviour in transitional regimes, where the current linear $(1 - p_{\text{bear}})$ scaling produces intermediate weights that may not reflect the true asymmetry of expected outcomes.

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A. Full Sensitivity and Ratio Table

Table 8: Full Sensitivity Analysis — All Configurations, Both Floors, All Investor Types

Sleeve	Investor	CAGR	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
Score improvement floor = 0.001								
20%	MV	8.06%	0.923	−26.78%	7.05%	1.159	0.301	32.6%
20%	MVS_cons	8.09%	0.927	−27.00%	6.94%	1.181	0.300	30.7%
20%	MVS★	8.10%	0.927	−27.00%	6.95%	1.181	0.300	30.7%
20%	MVK	8.04%	0.923	−27.38%	6.93%	1.176	0.294	29.2%
30%	MV	8.33%	0.935	−25.39%	7.30%	1.156	0.328	8.3%
30%	MVS_cons	8.32%	0.936	−25.51%	7.23%	1.166	0.326	8.0%
30%	MVS★	8.39%	0.943	−25.62%	7.24%	1.173	0.327	8.3%
30%	MVK	8.24%	0.923	−26.31%	7.26%	1.150	0.313	7.2%
35%	MV	8.55%	0.944	−25.82%	7.42%	1.167	0.331	4.5%
35%	MVS_cons	8.49%	0.941	−25.82%	7.40%	1.162	0.329	5.7%
35%	MVS★	8.58%	0.949	−25.82%	7.36%	1.179	0.332	5.7%
35%	MVK	8.39%	0.930	−25.91%	7.41%	1.148	0.324	6.8%
40%	MV	8.57%	0.931	−26.54%	7.58%	1.147	0.323	3.0%
40%	MVS_cons	8.67%	0.945	−26.54%	7.59%	1.158	0.327	4.2%
40%	MVS★	8.74%	0.952	−26.54%	7.58%	1.167	0.330	4.5%
40%	MVK	8.65%	0.944	−26.75%	7.58%	1.154	0.323	4.9%
45%	MV	8.60%	0.920	−27.25%	7.78%	1.123	0.316	1.5%
45%	MVS_cons	8.78%	0.939	−27.25%	7.77%	1.145	0.322	3.0%
45%	MVS★†	8.90%	0.956	−27.25%	7.72%	1.167	0.327	3.4%
45%	MVK	8.76%	0.941	−27.64%	7.76%	1.145	0.317	3.0%
50%	MV	8.73%	0.923	−27.95%	7.91%	1.120	0.312	0.8%
50%	MVS_cons	8.79%	0.932	−28.15%	7.93%	1.125	0.312	1.9%
50%	MVS★	8.93%	0.945	−28.34%	7.91%	1.145	0.315	1.5%
50%	MVK	8.71%	0.920	−28.36%	7.96%	1.112	0.307	1.5%
Score improvement floor = 0.002								
20%	MV★	7.85%	0.909	−27.33%	6.86%	1.162	0.287	72.7%
20%	MVS_cons	7.76%	0.898	−27.55%	6.83%	1.155	0.282	73.1%
20%	MVS	7.75%	0.897	−27.55%	6.86%	1.148	0.281	73.5%
20%	MVK	7.80%	0.901	−27.92%	6.88%	1.152	0.280	70.5%
30%	MV	8.02%	0.921	−25.39%	7.10%	1.145	0.316	58.7%
30%	MVS_cons★	8.16%	0.934	−25.51%	7.06%	1.170	0.320	58.3%
30%	MVS	8.10%	0.929	−25.62%	7.07%	1.161	0.316	58.3%
30%	MVK	8.18%	0.933	−26.31%	7.06%	1.173	0.311	53.8%
35%	MV	8.22%	0.933	−25.28%	7.22%	1.153	0.325	48.5%
35%	MVS_cons★	8.40%	0.950	−25.28%	7.20%	1.180	0.332	47.7%
35%	MVS	8.38%	0.946	−25.55%	7.20%	1.178	0.328	47.3%
35%	MVK	8.43%	0.946	−25.90%	7.28%	1.171	0.325	44.3%

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Sleeve	Investor	CAGR	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
40%	MV	8.20%	0.916	−25.99%	7.52%	1.107	0.315	40.2%
40%	MVS_cons	8.43%	0.938	−26.46%	7.50%	1.139	0.319	37.9%
40%	MVS	8.45%	0.939	−26.46%	7.49%	1.142	0.319	37.5%
40%	MVK★	8.54%	0.953	−26.75%	7.39%	1.168	0.319	35.2%
45%	MV	8.32%	0.912	−26.70%	7.61%	1.111	0.312	30.3%
45%	MVS_cons	8.55%	0.938	−27.16%	7.56%	1.145	0.315	28.8%
45%	MVS★	8.66%	0.952	−27.16%	7.52%	1.165	0.319	28.4%
45%	MVK	8.44%	0.928	−27.64%	7.55%	1.134	0.305	25.4%
50%	MV	8.56%	0.919	−27.86%	7.75%	1.121	0.307	23.1%
50%	MVS_cons	8.67%	0.933	−28.15%	7.79%	1.129	0.308	19.7%
50%	MVS★	8.69%	0.936	−28.34%	7.75%	1.137	0.307	19.3%
50%	MVK	8.74%	0.934	−28.36%	7.82%	1.133	0.308	17.8%

Notes: Benchmark (60/40): CAGR = 7.71%, Sharpe = 0.875, Max DD = −30.61%, Downside Dev = 7.11%, Sortino = 1.104, Calmar = 0.252. All configurations: rolling 60-month window, 60/40 core, 5 bp transaction costs, 2004–2026, up to two simultaneous satellite tilts. ★ = best Sharpe within the floor/sleeve group. † = global best Sharpe (0.956) across all configurations. **Bold rows** = primary configuration (45%/0.001).